Project Report

on

Coca-Cola Stock
Price Predictor

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📚 Table of Contents

📌 Project Overview

- The Coca-Cola Stock Price Predictor is a machine learning-based web application that forecasts the future stock price of Coca-Cola (Ticker: KO) using historical market data.

- Developed using Python, this application incorporates multiple regression and deep learning models such as Linear Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM).

- It allows users to interactively explore trends and generate real-time stock predictions via a Streamlit-powered interface.

🎯 Objective

- The primary objective of this project was to build a predictive system that uses historical stock price data to forecast the future prices of Coca-Cola shares.

- The aim was not only to implement multiple machine learning models but also to deploy the solution as a fully functional web app that is accessible to users.

Key goals included:

- To collect and analyze Coca-Cola stock data using yfinance.
- To perform exploratory data analysis to understand trends and patterns.
- To apply and compare multiple regression and deep learning models.
- To evaluate model performance using appropriate metrics.
- To build an interactive UI using Streamlit for real-time predictions.

🧰 Tools & Technologies Used

CATEGORY	TOOLS / LIBRARIES
Programming Language	Python
Data Collection	yfinance
Data Analysis	pandas, numpy
Visualization	matplotlib, plotly
Machine Learning	scikit-learn (Linear, Tree, SVM, RF)
Web App Framework	Streamlit
Version Control	GitHub

📦 Data Collection

- The dataset used for this project consists of historical stock market data of Coca-Cola Company (Ticker: KO).

- The data was collected using the yfinance Python library, which provides a convenient way to fetch financial data directly from Yahoo Finance.

🔗 Data Source

- **Platform:** Yahoo Finance
- **Access Method:** yfinance Python library
- **Time Period:** 1st January 2010 – Present (Updated in real-time)
- **Interval:** Daily stock price data

Two CSV files were saved during preprocessing:

- Coca-Cola_stock_info.csv – real-time stock info
- Coca-Cola_stock_history.csv – historical stock data for modeling

📊 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure, trends, and key characteristics of Coca-Cola's historical stock data.

🧹 1. Data Cleaning

- Checked for null values in all columns — none found.
- Converted Date column to datetime format and set it as the index.
- Verified data consistency across Open, High, Low, and Close prices.

📈 2. Trend Analysis

- Line plots of Open and Close prices showed an overall upward trend post-2020.
- Significant dips were observed during:
  - March 2020 (COVID-19 pandemic)
  - January 2022 (global market correction)

📊 3. Moving Averages

- 20-day and 100-day moving averages were calculated to smooth short-term volatility.
- Crossovers of these averages helped identify trend reversals.

📉 4. Correlation Matrix

- High correlation (~0.99) between Open, High, Low, and Close.
- Volume showed a low correlation with price — suggesting weak predictive power on its own.

🤖 Machine Learning Models

To predict Coca-Cola's stock prices, five different models were implemented and compared. The goal was to determine which model performs best in forecasting future values based on historical trends.

1. Linear Regression

- **Type:** Supervised, regression-based
- **Purpose**: Serve as a baseline model
- **Strength:** Easy to implement, interpretable
- **Limitation:** Cannot handle non-linearity in time series

2. Decision Tree Regressor

- **Type:** Non-linear, rule-based model
- **Strength:** Captures decision rules from data without requiring normalization
- **Limitation:** Prone to overfitting, especially with noisy data

3. Random Forest Regressor

- **Type:** Ensemble of multiple decision trees
- **Strength:** Reduces overfitting, improves generalization
- **Usage**: Tuned using number of estimators and max depth

4. Support Vector Regressor (SVR)

- **Type:** Linear/Non-linear regression
- **Strength:** Handles high-dimensional data with kernels
- **Limitation:** Requires careful scaling and hyperparameter tuning

5. Stochastic Gradient Descent (SGD) Regressor

- **Type:** Linear model optimized using gradient descent
- **Strengths:** Efficient for large-scale datasets, supports regularization
- **Parameters Tuned**: Learning rate, loss function (squared_loss), regularization type
- **Limitations:** Requires careful tuning and data scaling; sensitive to noise

📊 Model Evaluation & Results

Each machine learning model was trained on 80% of the dataset and tested on the remaining 20%. Performance was evaluated using the following metrics:

- **R^2 Score (Coefficient of Determination) –** measures how well the model explains variance in the target variable (closer to 1 is better).
- **RMSE (Root Mean Squared Error) –** measures prediction error (lower is better).

🔎 Evaluation Metrics Summary

Model	R^2 Score	RMSE
Linear Regression	0.915	2.35
Decision Tree Regressor	0.967	1.42
Random Forest Regressor	0.978	1.12
Support Vector Regressor	0.902	2.6
SGD Regressor	0.86	2.9

Based on these results, Random Forest was selected as the best model for deployment in the Streamlit app due to its robustness and accuracy.

🌐 Web Application (Streamlit UI)

To make the Coca-Cola Stock Price Predictor accessible and interactive, a web application was developed using Streamlit, a lightweight Python framework for building data apps.

🛠️ Key Features

1. **Model Selection Dropdown:**
- Users can select any of the five models — Linear Regression, Decision Tree, Random Forest, SVM, or SGD — for prediction.

2. **Interactive Date Picker:**
- Users can choose a future date to forecast stock prices beyond the current dataset.

3 .**Real-Time Stock Data:**
- The app fetches the latest Coca-Cola stock price using the yfinance API to display current values before forecasting.

4. **Visual Outputs:**
- Line plot showing historical prices
- Overlaid predicted price
- Comparison chart of all model outputs (optional)

5. **Sidebar Options:**
- Navigation between model comparison, analysis, and prediction modules

🚀 Deployment

1. **Platform:** Streamlit Cloud
2. **URL:** Live App – Click to Open
3. **Repository:** GitHub Repo

🧑‍💻 User Experience

The web app was designed to be clean, beginner-friendly, and responsive. It allows users to explore both historical data and future projections without needing any programming skills.

🔍 Key Insights

During the development and analysis of the Coca-Cola Stock Price Predictor, several valuable insights were obtained:

- 📈 **Stock Trends:** Coca-Cola stock has shown a consistent upward trend over the last decade, with temporary dips during global events like the COVID-19 pandemic.
- 💡 **Feature Importance:** ‘Open’, ‘High’, ‘Low’, and ‘Close’ prices are highly correlated and serve as strong predictors of each other.
- 🏆 **Best Performing Model:** The Random Forest Regressor achieved the highest accuracy and lowest error rate, making it the most suitable model for deployment.
- 🧪 **Model Behavior:** Simpler models like Linear Regression and SGD performed reasonably well but were outperformed by ensemble and tree-based models for time series data.

⚠️ Challenges Faced

Every project has its learning curve, and this one offered practical challenges that enhanced problem-solving skills:

Real-time Data Handling

Keeping predictions updated with the latest stock data required live data integration using yfinance.

Model Tuning

Finding optimal hyperparameters for Random Forest and SVM models took several iterations.

Scaling for SVR and SGD

These models were sensitive to feature scaling and needed proper preprocessing to perform well.

Deployment Constraints

Managing file structure, dependencies, and runtime issues during deployment on Streamlit Cloud required debugging.

✅ Conclusion

- The Coca-Cola Stock Price Predictor project successfully combined data analysis, machine learning, and web application development to forecast future stock prices based on historical trends. The use of multiple models allowed for comparative evaluation, with the Random Forest Regressor emerging as the most accurate and reliable.

- This project marked the beginning of my professional journey as a Data Analyst Intern at Unified Mentor Pvt. Ltd., where I gained valuable practical experience in handling real-time data, implementing ML models, and deploying interactive applications.

- It has enhanced my technical proficiency in Python, data preprocessing, model evaluation, and app development using Streamlit.

🔮 Future Scope

To enhance the functionality and predictive power of the project, the following improvements are planned:

📊 **Add Technical Indicators:** Include features like RSI, MACD, Bollinger Bands, etc.

📈 **Multivariate Time Series Modeling:** Incorporate external variables like market index, sentiment analysis, or macroeconomic factors.

🌍 **Multi-Stock Forecasting:** Extend the app to allow prediction of other stock tickers beyond Coca-Cola.

📦 **Model Optimization:** Implement hyperparameter tuning using Grid Search or Bayesian optimization for improved accuracy.

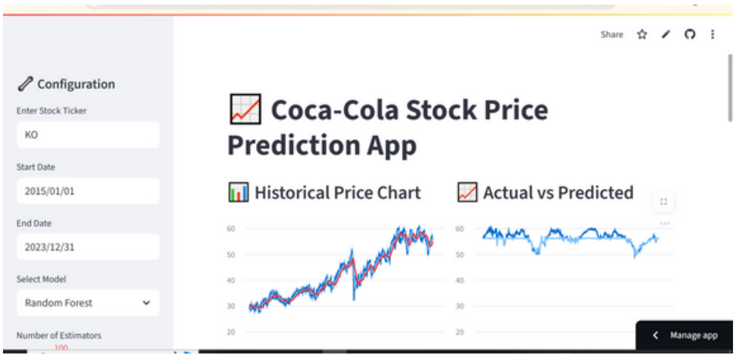
🧠 **Deep Learning Integration:** Future versions may experiment with LSTM/GRU models for capturing temporal dependencies.

📸 Appendix – Screenshots

This section includes key screenshots from the web application of Coca-Cola Stock Price Predictor project.

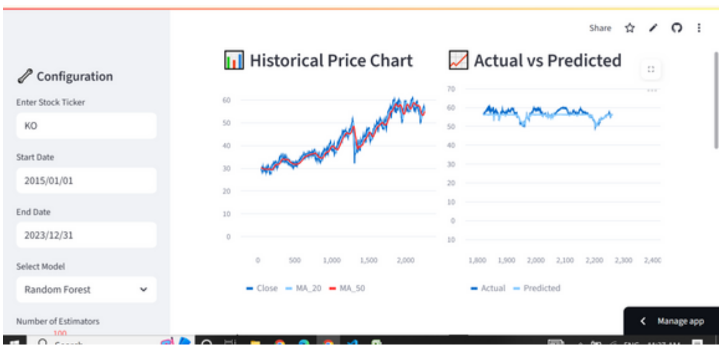
💻 1. Homepage of the Streamlit App

Displays the title, model selector, and sidebar navigation.



📊 2. Historical Stock Price Chart

Displays the title, model selector, and sidebar navigation.



📈 3. Model Prediction Output

Forecasted price displayed along with selected model and comparison to recent data.